MIT 18.05 — Complete Exam Preparation: Classes 22, 23 and 26

This guide follows the official MIT 18.05 materials, preserving the class-by-class structure, intuition, derivations, worked examples, and exam exercises.

Class	Topic	Main question
22	Confidence intervals for normal data	How can we estimate an unknown population mean?
23	Three views of confidence intervals; non-normal data	Why do confidence intervals work, and when can we trust them?
26	Linear and multiple regression	How can we explain and predict one variable using others?

These lectures connect with the probability material from Classes 1–8, especially expected value, variance, the normal distribution, the central limit theorem, covariance and correlation.

We build everything from first principles, then move to mathematical derivations, worked examples, and exam exercises.

Part I — Class 22: Confidence Intervals

1. The fundamental problem: knowing a population from a sample

Imagine a company employs 2,000 engineers. You want to know their average monthly salary.

The true average is called the **population mean**, denoted by μ.

You cannot easily ask all 2,000 employees, so you randomly select 100 and calculate their average salary.

Suppose you obtain:

$$\bar{X} = 15{,}000 \text{ MAD}$$

Here, $\bar{X}$ is the **sample mean**.

But does that mean the actual average salary of all employees is exactly 15,000 MAD?

Almost certainly not.

If you randomly selected another 100 engineers, perhaps you would get 14,700 MAD, or 15,200 MAD.

This gives us the essential problem of statistical inference:

We have an unknown fixed quantity μ, but our estimate $\bar{X}$ is random because the sample is random.

A confidence interval is a systematic way of expressing how much uncertainty surrounds our estimate.

For example, we might report:

$$\boxed{14{,}600 \leq \mu \leq 15{,}400}$$

using a procedure with a 95% confidence level.

The important question is: **Where do those two boundaries come from mathematically?**

That's what Class 22 teaches.

2. The normal distribution and why it appears here

Before constructing confidence intervals, recall the central limit theorem.

Suppose:

$$X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$$

This means each observation follows a normal distribution, with:

- μ: true population mean.
- σ^2: population variance (a measure of spread).
- σ: population standard deviation.
- n: number of independent observations.

The sample mean is:

$$\bar{X} = \frac{X_1 + \cdots + X_n}{n}$$

Using the properties of expectation and variance:

$$E[\bar{X}] = \mu$$

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

Therefore:

$$\boxed{\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)}$$

This is exact when the observations are independent and normal.

Why do we divide variance by n?

Imagine rolling a die.

One roll can be far from the expected value of 3.5. But if you roll a die 10,000 times, the average becomes much more stable.

The random positive and negative deviations partly cancel each other.

Mathematically:

$$\text{Var}\left(\frac{1}{n}\sum X_i\right) = \frac{1}{n^2}\sum \text{Var}(X_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

Notice something important: the standard deviation of the sample mean is:

$$\boxed{\text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}}$$

This is called the **standard error**.

Don't confuse it with standard deviation:

Quantity	Meaning
Standard deviation σ	How much individuals differ from one another

Quantity	Meaning
Standard error $\sigma/\sqrt{n}$	How much sample averages differ from one sample to another

The square root of n is crucial. Taking four times as many observations reduces the standard error by half.

3. The Z-score: transforming our problem into a universal one

Suppose you know a value x, and the distribution has mean μ and standard deviation σ.

The standardized value is:

$$Z = \frac{X - \mu}{\sigma}$$

A Z-score tells you how many standard deviations a value lies above or below the mean.

For the sample mean, we must instead divide by its standard error:

$$\boxed{Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}}$$

This gives:

$$Z \sim N(0, 1)$$

The great advantage is that we can now use the same standard normal distribution for countless different problems.

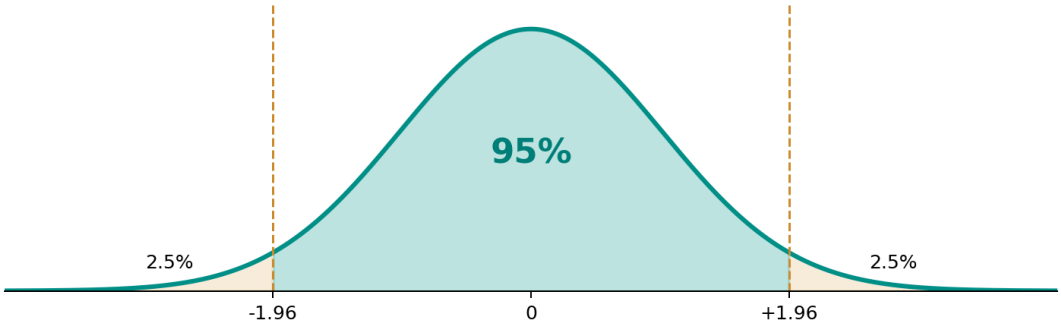


Figure 1: The middle 95% of a standard normal distribution lies between -1.96 and +1.96.

This means:

$$P(-1.96 \leq Z \leq 1.96) = 0.95$$

Why 1.96? It is the value that leaves 2.5% of the probability in each tail.

Other important values:

Confidence	α	Critical Z
90%	0.10	1.645
95%	0.05	1.960
99%	0.01	2.576

Here α is the total probability left outside the central interval.

4. Deriving the confidence interval formula

Now we combine everything.

Start with:

$$P\left(-1.96 \leq \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \leq 1.96\right) = 0.95$$

Multiply the inequality by $\sigma/\sqrt{n}$:

$$P\left(-1.96\frac{\sigma}{\sqrt{n}} \leq \bar{X}-\mu \leq 1.96\frac{\sigma}{\sqrt{n}}\right) = 0.95$$

Rearrange to isolate μ:

$$P\left(\bar{X}-1.96\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X}+1.96\frac{\sigma}{\sqrt{n}}\right) = 0.95$$

There it is!

95% Z confidence interval

$$\boxed{\bar{x} \pm 1.96\frac{\sigma}{\sqrt{n}}}$$

For a general confidence level $1-\alpha$, replace 1.96 with $z_{1-\alpha/2}$.

We haven't invented an interval. We started with an event that happens 95% of the time and used algebra to turn it into an interval containing the unknown mean.

Worked example 1 — Z confidence interval

A factory produces parts whose standard deviation in length is known to be 20 mm. We randomly inspect 64 parts.

Their average length is 100 mm.

Find the 95% confidence interval for the population mean.

Given:

$$n = 64, \quad \bar{x} = 100, \quad \sigma = 20$$

First calculate the standard error:

$$SE = \frac{20}{\sqrt{64}} = \frac{20}{8} = 2.5$$

Next calculate the margin of error:

$$ME = 1.96(2.5) = 4.9$$

So:

$$CI = [100-4.9,\ 100+4.9]$$

$$\boxed{CI = [95.1,\ 104.9]\ \text{mm}}$$

This method produces intervals that contain the true mean in 95% of repeated samples under the model assumptions.

5. Why 95% does not mean the parameter has a 95% probability of being inside

This distinction is important enough to appear again in Class 23.

Imagine that the true population mean is exactly 100, but we don't know it.

We repeatedly draw samples and calculate 95% confidence intervals.

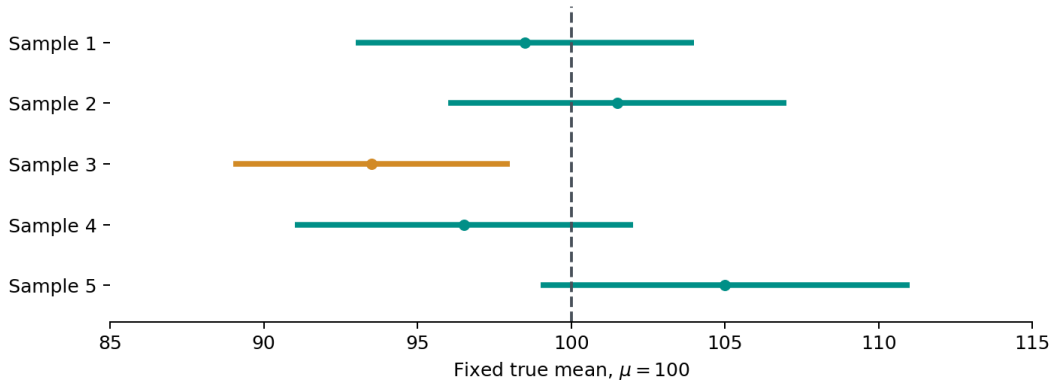


Figure 2: Illustrative repeated confidence intervals. The dashed line marks the fixed true mean.

The vertical line represents the true, fixed mean. Every horizontal line represents a new, random confidence interval.

Some intervals capture the true mean, and some miss.

A 95% confidence procedure captures the true mean on 95% of repetitions in the long run.

Once the sample has been observed, the particular interval either contains the fixed parameter or does not. The 95% describes the procedure's coverage, not a posterior probability for that specific interval.

6. Why increasing the confidence level makes an interval wider

Think of throwing a net into a lake to catch a particular fish.

If you use a wider net, you have a better chance of catching it.

But you lose precision.

The same happens here:

$$ME = z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}$$

Increasing the confidence level increases the critical value, increasing the width of the interval.

With $\bar{x} = 100$, $\sigma = 20$, $n = 64$:

Confidence	Interval
90%	[95.89, 104.11]
95%	[95.10, 104.90]
99%	[93.56, 106.44]

There is a trade-off between confidence and precision.

7. What if the population standard deviation is unknown?

Until now, we've assumed that σ is known.

In real problems, that often isn't true.

You might have 16 employee salaries, but no knowledge of the true variation across the entire company.

We can estimate standard deviation from the sample:

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

Why $n-1$, rather than n?

Because we've already estimated the sample mean from those observations. Once you know the mean and the first $n-1$ data values, the final value is constrained by their sum.

The sample deviations satisfy:

$$\sum_{i=1}^{n}(x_i - \bar{x}) = 0$$

So only $n-1$ deviations are free to vary.

This explains the **degrees of freedom** and helps motivate the $n-1$ denominator used for the unbiased sample variance.

But there is a new problem: if we replace σ with s, we introduce additional randomness.

So the standardized statistic is no longer exactly normal for small normal samples.

It instead follows the **Student's t-distribution**.

Intuition behind Student's t-distribution

Student's t-distribution looks like a normal curve, but with heavier tails.

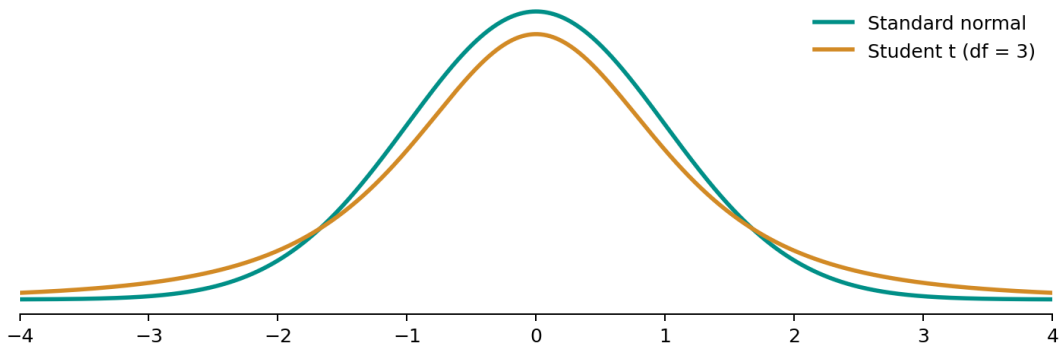


Figure 3: Student t with three degrees of freedom compared against the standard normal distribution.

Why heavier tails?

Because we are not only uncertain about the mean. We are also estimating how variable the original population is.

Sometimes the sample standard deviation is unusually small, making our standardized statistic unusually large in magnitude.

The t-distribution accounts for this.

As the sample size increases, estimating σ becomes more reliable, and the t-distribution approaches the normal distribution.

Formula for the t confidence interval

When the original observations are independent and normally distributed, with unknown σ:

$$\boxed{CI = \bar{x} \pm t_{1-\alpha/2,\ n-1}\frac{s}{\sqrt{n}}}$$

The critical value depends on $n-1$ degrees of freedom.

For example, a 95% t interval uses:

Degrees of freedom	t critical value
4	2.776
9	2.262
15	2.131
29	2.045
Very large	Approaches 1.960

Worked example 2 — t confidence interval

We collect 16 exam scores, obtaining:

$$\bar{x} = 80, \quad s = 12, \quad n = 16$$

Find a 95% confidence interval for the true mean score, assuming independent normal observations.

The population standard deviation is unknown, so use t.

Degrees of freedom:

$$df = n - 1 = 15$$

From the t table:

$$t_{0.975,15} = 2.131$$

Standard error:

$$SE = \frac{12}{\sqrt{16}} = 3$$

Margin of error:

$$ME = 2.131(3) = 6.393$$

Therefore:

$$\boxed{CI = [73.61,\ 86.39]}$$

If we incorrectly used $z = 1.96$, our interval would be narrower and would fail to account properly for the uncertainty in the estimated standard deviation.

8. Z versus t: the important decision

Situation	Method
Normal population, known σ	Z confidence interval
Normal population, unknown σ	t confidence interval

Situation	Method
Non-normal population, large sample	Approximate normal confidence interval using CLT
Estimating variance for normal data	Chi-square confidence interval

When the sample is small and the data are not approximately normal, neither the normal-data t procedure nor a large-sample Z approximation is automatically reliable.

9. Chi-square confidence intervals: estimating variance instead of the mean

Here is another major part of Class 22 that is easy to overlook.

Until now, our unknown quantity was μ, the population mean.

But sometimes we care more about consistency than the average.

Suppose two machines produce bolts with average lengths of exactly 10 cm.

Machine A produces lengths tightly concentrated around 10 cm.

Machine B sometimes produces 8 cm bolts and sometimes 12 cm bolts.

Same average, very different quality!

We therefore want to estimate:

$$\sigma^2$$

the population variance.

Why the chi-square distribution?

For independent normal observations:

$$\boxed{\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}}$$

The chi-square distribution is nonnegative and generally asymmetric.

That makes sense: variance cannot be negative.

The statistic compares sample variation to true population variation.

If the sample variance is much larger than the population variance, the ratio becomes large; if it is much smaller, the ratio becomes small.

Using the central region of the chi-square distribution and rearranging the inequalities gives:

$$\boxed{CI_{\sigma^2} = \left[\frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}, \frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}}\right]}$$

Here the subscripts indicate *left-tail* quantiles. For example, $\chi^2_{0.975,9}$ is the value below which 97.5% of a chi-square distribution with 9 degrees of freedom lies.

Notice that the larger critical value goes in the denominator of the lower bound.

Worked example 3 — Variance interval

A sample of 10 normal observations has:

$$n = 10, \quad s^2 = 16$$

For a 95% interval and 9 degrees of freedom:

$$\chi^2_{0.975,9} = 19.023$$

$$\chi^2_{0.025,9} = 2.700$$

Calculate:

$$L = \frac{9(16)}{19.023} \approx 7.57$$

$$U = \frac{9(16)}{2.700} \approx 53.33$$

Therefore:

$$\boxed{CI_{\sigma^2} \approx [7.57,\ 53.33]}$$

This interval is asymmetric, unlike the Z and t intervals for the mean.

To obtain a confidence interval for the standard deviation itself, take the square root of both endpoints:

$$CI_{\sigma} \approx [2.75,\ 7.30]$$

10. Relationship to hypothesis testing

MIT explicitly connects confidence intervals with null hypothesis significance testing (NHST).

Suppose a company claims:

$$H_0 : \mu = 105$$

But our observed 95% confidence interval is:

$$[95.1,\ 104.9]$$

Since 105 is outside the interval, the corresponding two-sided Z-test rejects that null hypothesis at a 5% significance level.

If the company claimed $\mu = 102$, we would not reject the hypothesis, because 102 lies inside the interval.

For matching two-sided tests:

$$\boxed{\mu_0 \notin CI_{95\%} \iff \text{Reject } H_0 : \mu = \mu_0 \text{ at } \alpha = 0.05}$$

This doesn't mean that every value inside the interval is equally plausible, nor that failing to reject proves a hypothesis true.

Class 22: exam exercises and solutions

Exercise 1. A normal population has known $\sigma = 18$. A sample of 81 observations has mean 72. Find the 95% Z interval.

Solution. $SE = 18/9 = 2$. Margin $= 1.96 \times 2 = 3.92$. Interval $\boxed{[68.08, 75.92]}$.

Exercise 2. A normal sample has $n = 25$, mean 40 and sample standard deviation 10. Find the 95% t interval. Use $t_{0.975,24} = 2.064$.

Solution. $SE = 10/5 = 2$. Margin $= 2.064 \times 2 = 4.128$. Interval $\boxed{[35.87, 44.13]}$.

Exercise 3. If a 95% interval is $[20, 30]$, would a matching two-sided test reject $H_0 : \mu = 32$ at $\alpha = 0.05$?

Solution. Yes. The hypothesized value 32 is outside the confidence interval.

Exercise 4. How many observations do you need for a 95% Z interval with margin of error at most 2, if $\sigma = 10$?

Solution. Require $1.96 \times 10/\sqrt{n} \leq 2$, giving $n \geq 96.04$. Round up: $\boxed{n = 97}$.

Part II — Class 23: Understanding Confidence Intervals Deeply

Class 23 extends Class 22 in two major ways:

1. It explains three mathematical perspectives on confidence intervals.
2. It shows how to construct intervals when the original population is not normally distributed.

The second point reconnects directly with the central limit theorem.

11. The three views of a confidence interval

MIT presents three complementary ways to understand confidence intervals.

View 1 — Standardized statistics

This is the approach we already used.

Start with a quantity whose distribution is known, such as:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

We know:

$$P(-1.96 \leq Z \leq 1.96) = 0.95$$

Rearranging produces:

$$\bar{X} \pm 1.96\frac{\sigma}{\sqrt{n}}$$

The main idea is:

Start with a probability statement involving the data, then rearrange it into a statement involving the unknown parameter.

This is often called the *pivot method*.

A pivot is a function involving observations and unknown parameters whose probability distribution is known without knowing those parameters.

For normal data we have three important pivots:

Pivot	Distribution
$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$	Standard normal
$\frac{\bar{X}-\mu}{S/\sqrt{n}}$	Student's t
$\frac{(n-1)S^2}{\sigma^2}$	Chi-square

All three allow us to construct confidence intervals through algebra.

View 2 — Inverting hypothesis tests

This perspective is especially interesting.

Imagine that someone proposes a particular value for the population mean.

For example:

$$H_0 : \mu = 100$$

You test it against your data.

Then another person proposes:

$$H_0 : \mu = 101$$

You test that too.

And then 102, 103, 104, and so on.

Instead of asking whether one particular value should be rejected, you ask:

Which hypothetical values of the parameter would not be rejected by my data?

The set of all such values forms a confidence set. In the usual one-parameter examples, it forms a confidence interval.

For instance, if all values from 95.1 to 104.9 are not rejected by the matching two-sided tests at $\alpha = 0.05$, the 95% confidence interval is:

$$[95.1, 104.9]$$

This is why the interval and the hypothesis test give consistent decisions.

The connection is:

$$\text{Confidence level} = 1 - \alpha$$

But this exact correspondence requires that the interval and hypothesis tests are constructed consistently.

View 3 — The formal definition

Let $I(X)$ be a confidence interval calculated from a random sample X.

Let θ represent any unknown parameter, such as a mean, variance or proportion.

An exact confidence interval procedure of level $1 - \alpha$ satisfies:

$$\boxed{P_\theta(\theta \in I(X)) = 1 - \alpha}$$

for every possible true value of θ in the model.

In simpler words:

No matter which permissible value of the parameter is the truth, the interval-building procedure captures that value with the stated long-run frequency.

Some procedures are **conservative**, meaning their coverage is at least $1 - \alpha$, rather than exactly equal to it.

This formal definition matters because it tells us what a confidence interval must guarantee, rather than simply prescribing one formula.

12. Confidence intervals versus Bayesian probability intervals

There is an important conceptual difference.

In frequentist statistics, the population parameter is fixed but unknown, and the interval is random before the sample is observed.

In Bayesian statistics, we represent uncertainty about the parameter using a probability distribution called the *posterior distribution*.

For example:

- A frequentist 95% confidence interval has 95% long-run coverage under its assumptions.
- A Bayesian 95% credible interval contains 95% of the posterior probability, conditional on the observed data, model and prior.

The numerical intervals may sometimes be identical, even though their interpretations differ.

MIT explicitly emphasizes this distinction.

13. Confidence intervals for proportions — Bernoulli data

Now suppose the original population is not normal.

Imagine asking randomly selected engineers:

"Have you previously participated in a programming competition?"

Each answer is either yes or no.

Define:

$$X_i = \begin{cases} 1 & \text{Yes} \\ 0 & \text{No} \end{cases}$$

Then:

$$X_i \sim \text{Bernoulli}(p)$$

Here p is the unknown proportion of engineers who answer yes.

For Bernoulli random variables:

$$E[X_i] = p$$

$$\text{Var}(X_i) = p(1-p)$$

The sample mean is:

$$\bar{X} = \frac{X_1 + \cdots + X_n}{n}$$

But because each X_i is either zero or one, $\bar{X}$ is simply the proportion of yes answers:

$$\boxed{\hat{p} = \frac{\text{Number of yes answers}}{n}}$$

We use $\hat{p}$ to estimate p.

Why can we use a normal approximation?

The individual answers are not normally distributed. They take only two values.

However, for sufficiently large n, the central limit theorem tells us that the sample proportion is approximately normal:

$$\hat{p} \approx N\left(p, \frac{p(1-p)}{n}\right)$$

Therefore:

$$\frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \approx N(0,1)$$

The main difficulty is that we don't know p, so we don't know the true standard error.

One common large-sample approximation is to substitute $\hat{p}$:

$$\boxed{CI_p \approx \hat{p} \pm 1.96\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}$$

This is called the Wald interval. It is simple, but can perform poorly for small samples or proportions near zero or one.

14. MIT's conservative confidence interval trick

The MIT reading teaches a useful alternative.

Recall:

$$\text{Var}(X) = p(1-p)$$

This quantity is maximized at $p = 0.5$.

To see this, differentiate:

$$f(p) = p - p^2$$

$$f'(p) = 1 - 2p$$

Setting the derivative to zero gives:

$$p = \frac{1}{2}$$

And therefore:

$$p(1-p) \leq \frac{1}{4}$$

Taking square roots:

$$\sqrt{p(1-p)} \leq \frac{1}{2}$$

Consequently, the standard error of a sample proportion satisfies:

$$SE(\hat{p}) \leq \frac{1}{2\sqrt{n}}$$

So we can use a worst-case approximation.

For a nominal 95% confidence interval:

$$CI \approx \hat{p} \pm 1.96\frac{1}{2\sqrt{n}}$$

Because 1.96 is approximately 2:

$$\boxed{CI \approx \hat{p} \pm \frac{1}{\sqrt{n}}}$$

This is MIT's **95% rule-of-thumb confidence interval**.

It is conservative relative to the normal approximation because we use the largest possible Bernoulli standard deviation.

The coverage guarantee remains approximate because we still rely on the CLT.

Worked example 4 — Survey of 400 people

A survey of 400 people finds that 224 prefer a particular product.

Calculate a 95% rule-of-thumb confidence interval for the true preference proportion.

First:

$$\hat{p} = \frac{224}{400} = 0.56$$

Then:

$$ME = \frac{1}{\sqrt{400}} = \frac{1}{20} = 0.05$$

So:

$$\boxed{CI = [0.51, 0.61]}$$

Or 51% to 61%.

The survey estimates that 56% of the population prefers the product, with a 5-percentage-point margin of error under the approximation.

An excellent exam question: Why is the margin of error proportional to $1/\sqrt{n}$?

Because averaging independent observations reduces the variance by a factor of n.

Standard deviation is the square root of variance, so uncertainty decreases as $1/\sqrt{n}$.

That gives an extremely useful rule:

- To reduce the margin of error by half, multiply your sample size by 4.
- To reduce it by a factor of 3, multiply your sample size by 9.
- To reduce it by a factor of 10, multiply your sample size by 100.

This is why increasingly precise surveys can become expensive.

15. Confidence intervals for general, non-normal data

We now generalize beyond the Bernoulli distribution.

Suppose individual observations follow a skewed distribution, such as an exponential distribution.

This might model certain waiting times.

Individual waiting times could be heavily skewed: most are short, but a few are very long.

However, the CLT says that for independent, identically distributed observations with finite positive variance, as n increases:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$$

Replacing the unknown population standard deviation by the sample standard deviation also gives an asymptotically standard-normal statistic.

This allows the large-sample confidence interval:

$$\boxed{CI_\mu \approx \bar{x} \pm 1.96\frac{s}{\sqrt{n}}}$$

Worked example 5 — Non-normal observations

Suppose 144 observations have:

$$\bar{x} = 50, \quad s = 12$$

The individual observations follow a non-normal distribution, but the sample is sufficiently large for the CLT approximation to be appropriate.

Calculate:

$$SE = \frac{12}{\sqrt{144}} = 1$$

The margin is:

$$ME = 1.96(1) = 1.96$$

Therefore:

$$\boxed{CI = [48.04, 51.96]}$$

How large must n be?

There is no universal answer.

For a symmetric, moderately behaved population, the normal approximation may become accurate fairly quickly.

For a highly skewed population with large outliers, it may need many more observations.

The MIT reading demonstrates this using simulations on exponential data. At $n = 20$, a nominal 95% large-sample interval covered the true mean in only about 90.5% of its trials; at $n = 400$, coverage rose to about 94.7%.

The lesson is that **sample size alone does not guarantee a good normal approximation; the underlying distribution matters.**

16. Class 23 exercises

Exercise 1. In a survey of 625 people, 350 answer yes. Find the 95% rule-of-thumb confidence interval.

Solution. $\hat{p} = 350/625 = 0.56$. Margin $= 1/\sqrt{625} = 0.04$. Interval $\boxed{[0.52, 0.60]}$.

Exercise 2. Why is the rule-of-thumb interval called conservative?

Solution. Because $p(1-p) \leq 1/4$ for every Bernoulli parameter. Using $\sigma = 1/2$ produces an interval at least as wide as the corresponding normal approximation using the true standard deviation.

Exercise 3. A sample has 100 non-normal observations, mean 12, and sample standard deviation 5. Find the approximate 95% interval if the CLT is applicable.

Solution. $SE = 5/10 = 0.5$. Margin $= 1.96 \times 0.5 = 0.98$. Interval $\boxed{[11.02, 12.98]}$.

Exercise 4. True or false: After calculating a 95% frequentist confidence interval, the probability that the parameter lies inside that realized interval is exactly 95%.

Solution. False. The 95% describes long-run coverage of the random interval procedure, not posterior probability for a fixed parameter given the observed sample.

Part III — Class 26: Linear Regression

This lecture is especially useful because it connects probability, optimization, calculus, geometry and prediction.

The official lecture includes least squares, residuals, model assumptions, polynomial fitting, overfitting, regression to the mean and multiple regression.

17. The fundamental question: Can one variable help predict another?

Imagine you collect the number of hours students spend studying and their exam scores.

You obtain these observations:

Study hours x	Exam score y
1	40
2	50
3	65
4	70
5	85

We suspect that more studying is associated with higher scores.

But the relationship isn't exact.

Someone could study for five hours and still get a low score, because other factors matter.

We therefore model the exam score as:

$$\boxed{Y = aX + b + \varepsilon}$$

where:

- X: explanatory or predictor variable (hours studied).
- Y: response variable (exam score).
- a: slope, telling us the expected score change associated with one additional study hour.
- b: intercept, the predicted score when $X = 0$.
- ε: random error, representing effects our model doesn't explain.

The important thing is that we are not claiming every observation lies exactly on one line.

Rather, the observations fluctuate around an underlying relationship.

18. The idea of the best-fitting line

Imagine drawing a line through a cloud of points.

Which line is best?

You could try many different lines, but how do you compare them objectively?

We need a quantity that measures prediction errors.

If our predicted value for observation i is:

$$\hat{y}_i = ax_i + b$$

then its **residual** is:

$$e_i = y_i - \hat{y}_i$$

A positive residual means the actual value is above the fitted line.

A negative residual means the actual value is below it.

We might initially consider adding all residuals:

$$\sum e_i$$

But this is not useful because positive and negative residuals can cancel.

Instead, square them:

$$e_i^2 = (y_i - \hat{y}_i)^2$$

Then add the squared residuals:

$$S(a,b) = \sum_{i=1}^{n} (y_i - ax_i - b)^2$$

This is the **residual sum of squares**, also called RSS or SSE.

The method of least squares chooses a and b that minimize this quantity:

$$\boxed{(\hat{a}, \hat{b}) = \arg\min_{a,b} \sum_{i=1}^{n} (y_i - ax_i - b)^2}$$

This is a mathematical optimization problem!

19. Deriving the best-fitting line using calculus

This part should be familiar from multivariable calculus.

We want to minimize:

$$S(a,b) = \sum_{i=1}^{n} (y_i - ax_i - b)^2$$

Take the partial derivative with respect to b:

$$\frac{\partial S}{\partial b} = -2 \sum_{i=1}^{n} (y_i - ax_i - b)$$

Set it to zero:

$$\sum (y_i - ax_i - b) = 0$$

$$n\bar{y} - an\bar{x} - nb = 0$$

So:

$$\boxed{b = \bar{y} - a\bar{x}}$$

This means the fitted line must pass through the point:

$$(\bar{x}, \bar{y})$$

Now differentiate with respect to a:

$$\frac{\partial S}{\partial a} = -2\sum x_i(y_i - ax_i - b)$$

Set this to zero and substitute $b = \bar{y} - a\bar{x}$.

After rearranging:

$$\sum(x_i - \bar{x})(y_i - \bar{y}) = a\sum(x_i - \bar{x})^2$$

Therefore:

$$\boxed{\hat{a} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}}$$

and:

$$\boxed{\hat{b} = \bar{y} - \hat{a}\bar{x}}$$

These are the two main simple linear regression formulas you need for the exam.

Connection with covariance and correlation from Class 7

Recall the sample covariance:

$$s_{xy} = \frac{1}{n-1}\sum(x_i - \bar{x})(y_i - \bar{y})$$

And the sample variance:

$$s_x^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$$

Therefore:

$$\boxed{\hat{a} = \frac{s_{xy}}{s_x^2}}$$

Using the correlation coefficient $r = s_{xy}/(s_x s_y)$, we also obtain:

$$\boxed{\hat{a} = r\frac{s_y}{s_x}}$$

This has a beautiful interpretation:

- If r is positive, the slope is positive.
- If r is negative, the slope is negative.
- If $r = 0$, the fitted simple-regression line has zero slope, provided the predictor varies.
- The magnitude of the slope depends on the measurement scales of X and Y, not only on correlation.

20. Worked example 6 — The MIT regression example

The official reading uses these three points:

$$(0,1), \quad (2,1), \quad (3,4)$$

Let's calculate the fitted line completely.

First compute the means:

$$\bar{x} = \frac{0+2+3}{3} = \frac{5}{3}$$

$$\bar{y} = \frac{1+1+4}{3} = 2$$

Now calculate:

$$\sum (x_i - \bar{x})^2 = \frac{14}{3}$$

and:

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = 4$$

Therefore:

$$\hat{a} = \frac{4}{14/3} = \frac{6}{7}$$

For the intercept:

$$\hat{b} = 2 - \frac{6}{7} \cdot \frac{5}{3} = \frac{4}{7}$$

Our best-fitting line is:

$$\boxed{\hat{y} = \frac{6}{7}x + \frac{4}{7}}$$

For example, when $x = 2$:

$$\hat{y} = \frac{6}{7}(2) + \frac{4}{7} = \frac{16}{7} \approx 2.286$$

But the actual value is 1.

So the residual is:

$$e = 1 - 2.286 = -1.286$$

The fitted line is the optimal *compromise*. It does not necessarily pass through every data point.

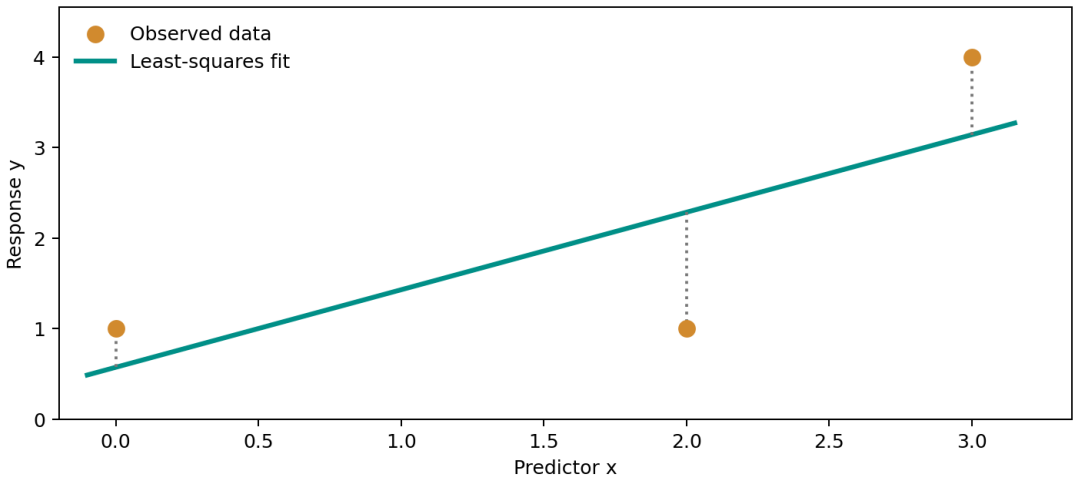


Figure 4: Observed values, least-squares line, and residuals.

Exam property: the residuals sum to zero

For ordinary least squares with an intercept:

$$\sum_{i=1}^{n} e_i = 0$$

This follows directly from differentiating the squared-error function with respect to the intercept.

Another useful consequence:

$$\sum x_i e_i = 0$$

The residuals are orthogonal to the intercept vector and predictor vector.

This is the beginning of the linear algebra interpretation of regression.

21. Residuals and homoscedasticity

The name is complicated, but the intuition is simple.

Homoscedasticity means that the error variance stays constant across predictor values.

For example, suppose the exam score fluctuations around the fitted line are approximately equally spread for students studying 2 hours and those studying 8 hours.

That's a homoscedastic pattern.

Heteroscedasticity means the error variance changes depending on the predictor.

For example, perhaps low study-hour students have scores tightly grouped together, while high study-hour students have extremely different results.

Then the residuals might create a fan-shaped pattern.

Why does it matter?

Classical linear regression inference often assumes independent errors with constant variance. Violating that assumption can make conventional uncertainty estimates unreliable.

Ordinary least squares can still calculate a fitted line, but confidence intervals and tests for its coefficients may need adjustments.

Other important assumptions

For the classical model, we generally want:

- A correctly specified linear relationship in the coefficients.

- Errors whose conditional mean is zero.
- Independent errors when the sampling setting requires independence.
- Constant error variance for conventional homoscedastic inference.
- Normally distributed errors for certain exact small-sample statistical tests.

Normal errors are not required simply to compute the least-squares coefficients.

22. Polynomial regression — Why can a curve still be called linear regression?

Suppose your data form a U-shaped pattern rather than a straight line.

You might fit:

$$y = ax^2 + bx + c$$

This is nonlinear in x, but linear in the unknown coefficients a, b, c.

That's the key distinction.

Linear regression means linear in the parameters, not necessarily linear in the input variable.

The squared-error function becomes:

$$S(a, b, c) = \sum_i (y_i - ax_i^2 - bx_i - c)^2$$

You minimize this over all three coefficients.

This is still a least-squares problem.

For example, the three MIT observations above lie exactly on:

$$\boxed{y = x^2 - 2x + 1}$$

Check:

- $x = 0 \Rightarrow y = 1$.
- $x = 2 \Rightarrow y = 1$.
- $x = 3 \Rightarrow y = 4$.

A quadratic model can fit those three observations perfectly.

But that raises another problem.

23. Overfitting — Why fitting the existing data perfectly can be bad

Imagine you have only 10 data points.

A sufficiently flexible polynomial can pass through every one of them.

Its training error might be zero.

Does that mean it is an excellent model?

No.

Your observations may contain random noise. If your model learns that noise instead of the underlying pattern, it may perform badly on future observations.

This is **overfitting**.

The fundamental trade-off is:

- A model that is too simple may miss the actual pattern (*underfitting*).
- A model that is too complex may chase accidental fluctuations (*overfitting*).

This connects regression to a central idea in machine learning: generalization to unseen data.

24. Measuring goodness of fit: R^2

The MIT slides introduce this concept, although they mark its detailed calculation as not required for MIT's own final exam. It is still useful to understand for an independent company exam.

Define total variation:

$$TSS = \sum (y_i - \bar{y})^2$$

This represents the squared prediction error if we always predict the sample mean.

Now define unexplained variation:

$$RSS = \sum (y_i - \hat{y}_i)^2$$

Then:

$$\boxed{R^2 = 1 - \frac{RSS}{TSS}}$$

For ordinary least squares with an intercept, R^2 tells us the fraction of observed variation in Y explained by the fitted model.

For example, if:

$$TSS = 100, \quad RSS = 20$$

then:

$$R^2 = 1 - \frac{20}{100} = 0.8$$

So the model explains 80% of the observed variation in the response in this dataset.

For simple linear regression with an intercept:

$$\boxed{R^2 = r^2}$$

where r is the sample Pearson correlation coefficient.

Notice that correlation has a sign, but R^2 does not.

A strongly negative linear relationship can have an R^2 close to 1.

25. Regression to the mean

This is one of the most interesting concepts in Class 26.

Suppose students take two noisy, positively correlated exams measuring similar abilities.

A student gets an exceptionally high score on the first exam.

What should we expect on the second?

The student may still perform above average, but we should generally expect their second score to be less extreme.

Why?

Because an unusually high observed score can be produced by two things:

1. High underlying ability.
2. A favorable random fluctuation.

That random fluctuation is unlikely to be equally favorable on the next exam.

This phenomenon is called **regression to the mean**.

For standardized variables in a linear regression setting, the predicted standardized response is:

$$\hat{z}_y = rz_x$$

If $r = 0.6$ and someone scores 2 standard deviations above average on the first measurement:

$$\hat{z}_y = 0.6(2) = 1.2$$

We predict that the second measurement will be 1.2 standard deviations above average, not 2.

This does not automatically indicate a decline in ability.

It can happen entirely because of imperfect correlation and random variation.

Important warning

Regression to the mean can create false conclusions about interventions.

For example, a company gives extra training only to employees who performed exceptionally badly on one test.

If those employees improve on the next test, some of the improvement may have occurred even without the intervention.

A controlled comparison is needed to establish whether the intervention caused the change.

26. Multiple linear regression

Now imagine predicting exam scores using more than one variable.

Perhaps the score depends on:

- Hours studied.
- Hours slept.
- Number of practice exams completed.

We could use:

$$\boxed{Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon}$$

This is multiple linear regression.

The logic is identical to simple regression: choose the coefficients that minimize the sum of squared residuals.

For example, suppose a fitted model is:

$$\hat{y} = 20 + 4x_1 + 2x_2$$

where x_1 is study hours and x_2 is sleep hours.

If a student studies for 5 hours and sleeps for 7:

$$\hat{y} = 20 + 4(5) + 2(7) = 54$$

The interpretation of $\beta_1 = 4$ is that, holding sleep hours constant, an additional study hour is associated with a 4-point increase in the model's predicted score.

The phrase **holding other variables constant** is critical.

Matrix notation (extra intuition)

If you remember linear algebra from CPGE, multiple regression can be written elegantly:

$$\mathbf{y} = X\beta + \varepsilon$$

Where:

- $\mathbf{y}$ is the vector of observed outputs.
- X is the design matrix containing the predictors and an intercept column.
- β is the vector of unknown coefficients.

We minimize:

$$\|X\beta - \mathbf{y}\|^2$$

Differentiation gives the normal equations:

$$X^T X\hat{\beta} = X^T\mathbf{y}$$

If $X^T X$ is invertible:

$$\boxed{\hat{\beta} = (X^T X)^{-1} X^T\mathbf{y}}$$

This is the same mathematics as simple linear regression, expressed in a more general form.

Geometrically, we are projecting the observed response vector onto the space spanned by the predictors.

In practical numerical software, QR or SVD methods are often preferable to explicitly calculating the inverse.

27. Class 26 exercises

Exercise 1. Fit a least-squares line to $(1, 2)$, $(2, 3)$, $(3, 5)$.

Solution. $\bar{x} = 2$, $\bar{y} = 10/3$. Sum of cross-deviations $= 3$, sum of squared x-deviations $= 2$. Slope $a = 3/2$, intercept $b = 1/3$. Thus $\boxed{\hat{y} = 1.5x + 1/3}$.

Exercise 2. A fitted regression line is $\hat{y} = 10 + 3x$. An observation has $x = 4$ and $y = 25$. What is its residual?

Solution. Predicted value $= 10 + 3 \times 4 = 22$. Residual $= 25 - 22 = \boxed{3}$.

Exercise 3. If $TSS = 120$ and $RSS = 30$, what is R^2?

Solution. $R^2 = 1 - 30/120 = \boxed{0.75}$. The model explains 75% of the observed variation in Y.

Exercise 4. If a simple regression has sample correlation $r = -0.8$, what is R^2?

Solution. $R^2 = (-0.8)^2 = \boxed{0.64}$. The negative correlation means the slope is negative, but $R^2 = 0.64$.

Exercise 5. Does a ninth-degree polynomial with zero training error necessarily predict future observations better than a quadratic model?

Solution. No. It may overfit random noise in the training data. Training accuracy and future prediction accuracy are different objectives.

Exercise 6. In a multiple regression, what does the coefficient of X_1 mean?

Solution. It represents the change in the predicted response for a one-unit increase in X_1, holding the other predictors constant.

Part IV — Final Exam Revision Sheet

These are the formulas and distinctions to prioritize for the exam.

Confidence interval formulas

Scenario	Formula
Normal data, known σ	$\bar{x} \pm z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}$
Normal data, unknown σ	$\bar{x} \pm t_{1-\alpha/2,n-1}\frac{s}{\sqrt{n}}$
Normal data, variance	$\left[\frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}, \frac{(n-1)s^2}{\chi^2_{\alpha/2}}\right]$, df $= n-1$
Bernoulli proportion, approximate	$\hat{p} \pm z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
Bernoulli, MIT conservative rule of thumb	$\hat{p} \pm \frac{1}{\sqrt{n}}$, approximately 95%
General mean, large sample	$\bar{x} \pm z_{1-\alpha/2}\frac{s}{\sqrt{n}}$

Regression formulas

$$\hat{y} = \hat{a}x + \hat{b}$$

$$\hat{a} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

$$\hat{b} = \bar{y} - \hat{a}\bar{x}$$

$$RSS = \sum(y_i - \hat{y}_i)^2$$

$$R^2 = 1 - \frac{RSS}{TSS}$$

How all these lectures connect with Classes 1–8

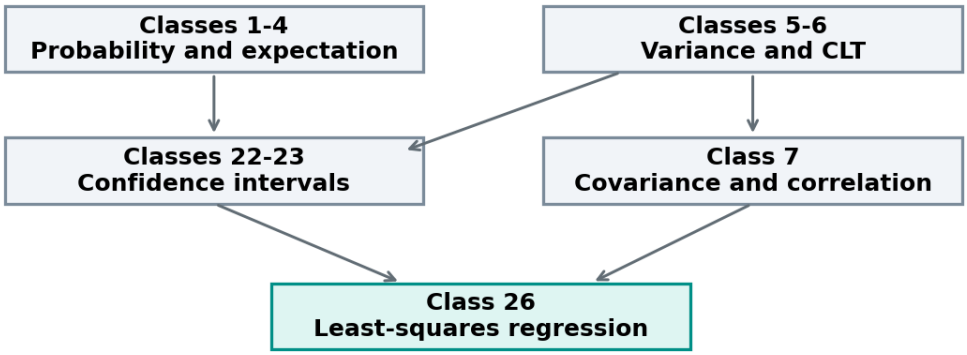


Figure 5: The early probability classes supply the foundation for inference and regression.

If you understand the *reason* behind each of these connections, the formulas become far easier to reconstruct during an exam.

Official MIT material for further practice

- MIT 18.05 official class materials
- Class 22 — Complete MIT reading (PDF)
- Class 23a — Three views of confidence intervals (PDF)
- Class 23b — Non-normal data (PDF)
- Class 26 — Complete MIT linear regression reading (PDF)

For the exam, aim for three levels of mastery: first, identify which mathematical method a question requires; second, carry out the computation correctly; third, explain why the method is valid and what its result means. The third level is especially important for conceptual questions about confidence, the CLT, or regression.

A timed mock exam covering exactly Classes 1–8, 22, 23 and 26 would combine calculations, short proofs, interpretation traps, and regression problems and help identify gaps before the company exam.